

$$\frac{\partial f}{\partial x} = yz$$

integrate in x

$$xyz + g(y, z)$$

$$\frac{\partial f}{\partial y} = xz$$

$\partial g / \partial y = 0$

$$xyz + h(z)$$

$$\frac{\partial f}{\partial z} = xy + 2z$$

$h'(z) = 2z$

$$xyz + z^2 + C$$